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Effect of online research data management instruction on social science graduate students' RDM skills

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ABSTRACT

Prior research has suggested the value of and the need to provide consistent research data management (RDM) instruction specifically for graduate students. However, there is a lack of RDM instruction that is tailored for the social science disciplines. The effect of a four-hour, online RDM instruction intervention, designed based on the research data life cycle, on the RDM knowledge of graduate students in social science disciplines was investigated. A total of 84 students completed both pre/post knowledge assessments with 40 students randomly assigned into the intervention group receiving online instruction and 44 in the control group. A one-way ANCOVA was used for the data analysis. Results indicated that social science graduate students who received online RDM instruction had a significantly higher score in RDM knowledge than students in the control group. Moreover, the effect of the instruction on participants' RDM skills varied by their disciplines.

1. Introduction

Research data management (RDM) consists of the actions a researcher takes to enhance the accessibility of data during the research project and to preserve data for use or sharing after a project is completed (Briney, 2015). Advances in technology have increased the types of digital data collected during research projects and made the ability to find, interpret, retain, protect, and preserve research data even more important (Briney, 2015). All researchers benefit from RDM, but graduate students as beginning researchers are a key audience for developing good habits in managing research data.

1.1. Problem statement

Despite the importance of RDM, graduate students receive limited RDM instruction (e.g., Pasek & Mayer, 2019; Phillips, Miller, & Miller, 2019; Xu, Zhou, Kogut, & Watts, 2022). When instruction is provided, the typical contexts are either a separate course or workshops typically taught by librarians (Schmidt & Holles, 2018a). Often these workshops focus on the needs of graduate students from multiple disciplines, not just the social sciences (Oo, Chew, Wong, Gladding, & Stenstrom, 2021; Xu et al., 2022). Social science researchers have unique needs when working with research data, like the use of secondary datasets and maintaining the confidentiality of participant data (Southall, 2018). In addition to a lack of instruction specifically for social science researchers, few studies have utilized intervention research designs to evaluate the effectiveness of instruction sessions on the development of students' RDM skills (Xu et al., 2022). More specifically, few studies have used statistical analysis to evaluate the effectiveness of RDM instruction designed for social science graduate students. This study aims to contribute to this research gap by using a one-way ANCOVA to examine the effectiveness of an online RDM instruction on students' RDM skills.

Therefore, this research study aims to fill in the gaps identified through the literature review by examining the effect of RDM instruction on graduate students using an experimental research design. The present study will add to the research base in three ways. First, the target population is graduate students in the social sciences. Second, the current study utilizes an experimental research design with randomized groups to determine the effectiveness of one type of instructional intervention: online learning. Finally, prior research experience and disciplinary background have been found to contribute to the development of RDM skills, so this study also focuses on those aspects within a college of education that includes distinct social sciences disciplines from teacher

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education and educational psychology to more science-focused health and kinesiology.

1.2. Research questions

For the current study, the authors conducted a four-hour online instruction about RDM with the graduate students in the School of Education and Human Development at Texas A&M University (TAMU) and explored the effectiveness of the online RDM instruction on the graduate students' RDM skills. This study addressed the following research questions:

1. What is the effect of the four-hour online RDM instruction on graduate students' RDM skills?
2. Does the effect of online instruction on graduate students' RDM skills vary by their disciplinary focus?
3. Does the effect of the online instruction on graduate students' RDM skills vary by their current year of the program?

2. Literature review

2.1. The importance of research data management training for graduate students

Over the last two decades, the demand for RDM services has grown in academic libraries at higher education institutions (Perrier, Blondal, & MacDonald, 2018). As RDM instruction plays a critical role in RDM services in academic libraries (Si, Xing, Zhuang, Hua, & Zhou, 2015), many academic libraries have established or adapted services to address the RDM needs of their target audiences across campus, including librarians, faculty, and researchers (Atwood et al., 2017; Johnson & Bresnahan, 2015; Williams, 2013). However, RDM instruction has remained a persistent gap that needs to be addressed (Goben & Griffin, 2019). In particular, there is a lack of appropriate training about RDM for graduate students (Xu, 2022; Xu et al., 2022).

Prior research has suggested the value and need for graduate students to be provided with consistent data management instruction (Johnston & Jeffryes, 2014; Raszewski et al., 2021; Williams, 2013). Knowledge and skills in RDM are necessary in all academic fields. Graduate students, however, seemed to not have adequate education in RDM, meaning that they were not well-prepared to manage data effectively (Carlson, Fosmire, Miller, & Nelson, 2011). For example, Phillips et al. (2019) investigated graduate students' knowledge about RDM at the Gumberg Library at Duquesne University. This study provided evidence that graduate students lack support with data analysis and data organization, and they needed hands-on and practical training in tools and technology to help in the RDM process.

In scientific disciplines, Pasek and Mayer (2019) examined graduate students and faculty members' knowledge in RDM and experience with RDM instruction. This study found that graduate students reported not being knowledgeable and skilled in most aspects of RDM. The study also showed that graduate students usually learned RDM-related skills by themselves, and the faculty members seemed to have little impact on graduate students' education in RDM. This finding is similar to Carlson, Johnston, Westra, and Nichols (2013), indicating that without formal training, graduate students develop RDM skills ad hoc and at the point of need by themselves.

2.2. Existing three models of RDM training for graduate students

Recently, more academic libraries have realized the importance of RDM training for graduate students, and some academic libraries in the United States have established services to support the development of graduate students' RDM skills (Tenopir et al., 2015). Literature has shown that there are three fundamental models in delivering RDM instruction to graduate students, including a) training workshops/seminars (Eaker, 2014; Fong & Wang, 2015), b) courses integrated with RDM instruction (Thielen & Hess, 2017), and c) formal RDM courses (Fong & Wang, 2015; Schmidt & Holles, 2018a; Thielen, Samuel, Carlson, & Moldwin, 2017).

For the RDM training workshops or seminars, one-time workshops and seminar series are often provided by academic libraries, which suggests that there is value in providing RDM training specifically for graduate students. For example, the one-day "Data Management Basics" Workshop introduced important aspects of managing data for graduate students in engineering and science disciplines and received positive feedback from participants (Eaker, 2014). Simultaneously, weekly seminar series about basic data management knowledge with more in-depth instruction and practices were also acknowledged for their significance in improving graduate students' RDM skills and meeting their needs (Fong & Wang, 2015).

As for course embedded RDM instruction, partnerships between academic libraries and a specific discipline or department are often seen in the literature. The RDM embedded instruction in courses aimed at helping graduate students understand basic knowledge about RDM and its importance in the research lifecycle. That is, collaboration between librarians and a disciplinary department that embeds RDM instruction into graduate-level classes is beneficial for introducing the graduate students to the concepts of RDM-related skills (Thielen & Hess, 2017).

For the formal course instruction in RDM skills, both non-credit courses and courses with credit-bearing structures have been described in the literature (Muilenburg, Lebow, & Rich, 2015; Pascuzzi & Nelson, 2018; Rantasaari & Kokkinen, 2019; Whitmire, 2015). Credit-based RDM instruction usually includes multiple stakeholders who garner support from a mix of professional backgrounds, such as research faculty, librarians, information technologists, legal experts, and/or data managers (Schmidt & Holles, 2018a; Thielen et al., 2017). Moreover, the credit-based course providing in-depth RDM knowledge for graduate students might be co-taught by a librarian and a faculty member with an active research program at the higher education institution which delivers broad knowledge on RDM tools and standards from the expertise of the librarian and relevant research-focused examples and experience from the faculty member (Schmidt & Holles, 2018a).

Non-credit programs for RDM instruction were also provided by some academic libraries (Griffin, 2020; Johnston & Jeffryes, 2015). Those courses were offered in-person to address basic data management skills and incorporated data information literacy competencies developed according to the identified graduate students' needs at various stages throughout the data life cycle. The feedback from their participants also showed a high level of satisfaction with the course and expressed an increase in their knowledge and skills in data management.

Besides these models, there are also some well-established open access online RDM learning platforms such as DataOne SkillBuilding Hub, MLA's Data Services Specialization, MANTRA, and RDMLA. However, these platforms do not target graduate students specifically. Moreover, these platforms require individuals to independently progress their learning, which requires high self-regulation and motivation for learners to finish the courses.

Despite the multiple models of RDM instruction, there is a lack of RDM instruction that is specifically tailored for the social science disciplines in the literature. RDM instruction for graduate students has focused on general skills through a basic introduction to RDM, while participants expressed willingness to receive more instruction and RDM resources specifically tailored for their unique discipline (Johnston & Jeffryes, 2014; Whitmire, 2015). According to existing studies, most of the discipline-tailored RDM instruction concentrated on STEM disciplines (Eaker, 2014; Fong & Wang, 2015; Frugoli, Etgen, & Kuhar, 2010; Schmidt & Holles, 2018a; Thielen et al., 2017). Only a few studies of RDM instruction focused on the social science disciplines (Akers & Doty, 2013; Xu et al., 2022). As social science disciplines heavily depend on both quantitative and qualitative research data, secondary data, and data from human subjects, which are distinct from some STEM

disciplines, it is crucial to develop and provide RDM instruction specifically for these subject areas (Thielen & Hess, 2017).

2.3. RDM skills vary across disciplines and research experience

Due to the varied nature of research data collected, the needs and skills of RDM are diverse across disciplines (Adamick, Reznik-Zellen, & Sheridan, 2012). For example, for data generated from engineering science, some is gathered through the work in the laboratory and fieldwork and may consist of images and spreadsheets, while researchers in social science usually use a mix of quantitative and qualitative data as well as secondary data collected by others (Fong & Wang, 2015; Thielen & Hess, 2017). Therefore, although many RDM skills are related across the disciplines, applying those skills might vary widely from discipline to discipline (Frugoli et al., 2010).

Research experience seems to be another factor that contributes to students' RDM skills. Doucette and Fyfe (2013) compared confidence in managing data between doctoral students and master's students from nine large research-based universities in Canada. They found that doctoral students had statistically significantly higher reported confidence levels than master's students. Increased experience as doctoral students research progressively in graduate studies might account for the higher confidence level than the master's students. Likewise, Gurat (2018) investigated the data management skills of college students at a private university using a mixed, qualitative and quantitative approach. The study found that senior students are more skillful in appropriately managing data than junior students.

2.4. Lack of intervention studies in the field

Another important characteristic of RDM instruction for graduate students is consideration of the ways that RDM instructional interventions have been assessed. In general, library and information science practitioners rely on case studies, conceptual studies based on existing literature, and questionnaires as research methods (Lund & Wang, 2021). Yet, researchers and organizations focused particularly on instruction have called for the use not only of a variety of methods beyond case studies (ACRL, 2021) but also for more advanced research methods including statistical analysis (Crawford & Feldt, 2007).

Although RDM research is still developing (Faniel & Connaway, 2018), existing reviews have highlighted the need for additional studies on the effectiveness of RDM interventions. In a scoping review looking broadly at RDM, Perrier et al. (2018) identified examining the impact of interventions as a future research direction after finding few studies that examined the effect of RDM interventions. In a systematic review focused on RDM training, Oo et al. (2021) found that one of four indicators of successful RDM training included "increased RDM knowledge and understanding, and increased confidence in enacting RDM practices" (p. 7). However, Oo et al. (2021) did not provide enough detail about how the increase in knowledge was determined in the analyzed studies to identify which research designs and data analysis methods were utilized. Examining the RDM instruction within academic libraries, Xu et al. (2022) found that 26.6% of the articles were needs assessments, while only 12.9% were intervention studies. Then, only four of the intervention studies used statistical analysis to determine the effect of RDM instruction (Xu et al., 2022).

Looking closely at the four intervention studies that utilized statistical analysis in the Xu et al. (2022) study, it is apparent that more studies examining online instructional interventions for graduate students are needed, as none of the studies evaluated the effect of a synchronous online RDM instruction session. The instruction modes were an experiential learning activity in an undergraduate course (Agogo & Anderson, 2019), online self-paced modules for librarians (Read, Larson, Gillespie, Oh, & Surkis, 2019), a three-hour credit course for graduate students (Schmidt & Holles, 2018a), and in-person workshops for librarians (Carlson, 2013).

3. Methods

3.1. Curriculum design

The authors of this study designed an intervention based on the research data life cycle–from data management plan to data sharing (Corti, Van den Eynden, Bishop, & Woollard, 2014; Federer, 2016; Whyte & Tedds, 2011). Texas A&M University Libraries started to provide a series of RDM-related workshops in Spring 2020, which received positive feedback from faculty, students, and researchers across campus. However, the Libraries never offered intensive RDM training targeting graduate students in one specific discipline. The present study paid special attention to the graduate students from the ignored RDM training discipline of social science (Akers & Doty, 2013). Based on the existing workshop curriculum, the authors selected those workshops that covered the basic elements of the research data life cycle, such as data management plan and data sharing (Corti et al., 2014; Federer, 2016; Whyte & Tedds, 2011) in the current intervention study. The intervention study was a part of the T3 grant project at TAMU, which aimed to equip the graduate students in the School of Education and Human Development (SEHD) with basic RDM skills through a four-hour intensive training about RDM.

The first session was an overview of RDM and a guide to best practices in RDM. This session introduced the definition of research data, RDM, and the research data life cycle. It also included strategies for file/folder management and offered examples from practical research. The second session was about data management plans. This session discussed the benefits of data management, and explained what a data management plan is, and navigated the graduate students through the DMPTool to develop a data management plan. The third session was about data sharing. In this session, instructors introduced the concept of FAIR data principles, encouraged the graduate students to explore data repository in their discipline, and demonstrated how to share data by using the Texas Data Repository. The fourth session was about metadata. In the last session, instructors introduced the advantages of data documentation and described how to optimize metadata to make the research data more findable, accessible, interoperable, and reusable. Each workshop name and its corresponding objectives are listed in Table 1.

In the workshop, instructors also used some active learning activities. The instructors used polls and breakout rooms to engage participants. The workshop also adopted some other strategies to attract participants' attention, such as encouraging them to turn on cameras, requiring them to take collective notes on a Google Doc, and Google Jamboard activities.

3.2. Research design

After receiving institutional review board approval, the authors used the institution's bulk email to recruit participants from SEHD. All the participants were graduate students and signed the consent form. A pre-survey was conducted through Qualtrics to collect the participants' demographic information and a pre-assessment was conducted to measure the participants' RDM skills. One hundred and fifty-five responses were received in the pre-test phase, of which 98 responses were valid.

Next, the authors randomly assigned the participants to the intervention or the control group. The intervention group participants received a four-hour instruction session about RDM through a Zoom session, which was offered twice to accommodate schedules. When the intervention was completed, the authors conducted the same knowledge assessment as the pre-assessment to measure participants' RDM skills. The authors shared the instructional materials with the participants of the control group when the study was completed. Eventually, eighty-four valid responses were received through Qualtrics in the post-test phase, in which 40 participants were from the intervention group while the other 44 were from the control group (Table 2). According to the participants' department affiliation and major program reported in

Table 1
Workshop and the learning objectives.

Workshop	Learning objectives
Research Data Management: A Guide to Good Practice	• Summarize their data workflow in order to consider a holistic data management plan • Compare file naming conventions in order to choose the appropriate data management strategy for their research needs • Recognize appropriate data storage practices in order to mitigate data loss during a research project
Creating Research Data Management Plans Using DMPTool	• Identify how data iterates throughout the data lifecycle in order to explain their research workflow in a DMP • Classify the basic elements of a data management plan in order to minimize data loss and maximize efficiency • Recognize data management, support and guidance in order to fully articulate a data management plan • Use DMPTool to develop a data management plan in order to apply for funding opportunities
Publishing Data in the Texas Data Repository	• Describe the characteristics of FAIR data in order to share data openly • Identify disciplinary data repositories in order to choose appropriate solutions for sharing data with colleagues • Use the Texas Data Repository in order to share data in an open access repository
Optimizing Metadata to Make Your Research Data Findable, Accessible, Interoperable, & Reusable	• Explain the value of documenting research data in order to include metadata in their research workflow • Define the different forms of metadata in order to choose the appropriate metadata strategy for their research context • Classify metadata into different categories in order to enable meaningful structure to their research data • Cite data appropriately in order to reuse existing data in their own research

Table 2
Demographic information of participants ($N = 84$).

Characteristics		N	%
Gender	Female	72	85.71%
	Male	10	11.90%
	Not indicated	2	2.38%
Race/Ethnicity	White	38	45.24%
	Asian	27	32.14%
	African American	5	5.95%
	Hispanic or Latino American	8	9.52%
	Native American	1	1.19%
	Pacific Islander	5	5.95%
Degree pursuing	Master level	13	15.48%
	Doctoral level	71	84.52%
Disciplinary focus	Educational Psychology (EPSY)	47	55.95%
	Teacher Education (TTED)	15	17.86%
	Educational Administration & Human Resource (EAHR)	8	9.52%
	Health & Kinesiology (HLKN)	14	16.67%

the questionnaire, four specific disciplines were identified, including Educational Psychology (EPSY), Teaching and Teacher Education (TTED), Educational Administration (EAHR), and Health and Kinesiology (HLKN). There was a three-months gap between pre- and post-assessment.

A prior power analysis indicated that a sample size of 44 was adequate for detecting the treatment effect when using ANCOVA analysis. In this power analysis, the power was set as 0.80, the probability of error α equal to 0.05, and the number of covariates was set by 3. The authors set the effect size as 0.44 since previous literature (Agogo & Anderson, 2019) indicated that a medium effect was expected due to experimental effects (Cohen's $d = 0.44$, two groups).

3.3. Measures

To measure participants' RDM skills, the authors designed a set of knowledge assessments about RDM based on the MANTRA Research Data Management Training course and Research Data Management Librarian Academy course's test, which contained 12 question items (See Appendix 1). The authors designed the assessment according to the Libraries' own RDM curriculum and learning objectives. Self-designed assessments have their own advantages and disadvantages (Johnson & Christensen, 2008; Xu, Chen, Eutsler, Geng, & Kogut, 2020).

3.4. Data analysis

3.4.1. Preliminary data analysis

Assumptions were checked prior to the data analysis for the research questions/hypotheses. First, no missing data and outliers were identified. Normality of the data was not violated and tested by the value of skewness (−0.13) /kurtosis (3.02), density plot, Q_Q plot (See Fig. 1), and Shapiro-Wilk tests ($p = .809$). The assumption of homogeneity was found to be tenable ($Chi^2 = 6.57$, $p = .16$) and the linearity assumption was met by a scatterplot of the residuals versus predicted values with a Lowess line showing little departure from the zero point (See Fig. 2).

3.4.2. Analytical plan

STATA 17 was used for the data analysis. Assumptions checking was performed prior to the main analysis. Descriptive statistics were used for presenting the pre/post scores of participants' RDM skills, including the number of participants, adjusted mean scores, and standard deviations in each group. The research questions were tested in order by a one-way ANCOVA. Eta-square was used for reporting the effect size.

For research question one, the research hypothesis was that there are statistically significant effects of online RDM instruction on graduate students' RDM skills, controlling for participants' pre-scores, discipline, and years of study in their current program. For research question two, the research hypothesis was that participants' RDM skills vary by their discipline, after controlling for the impact of their pre-scores and years of study in their current program. Follow-up mid- and micro assessments

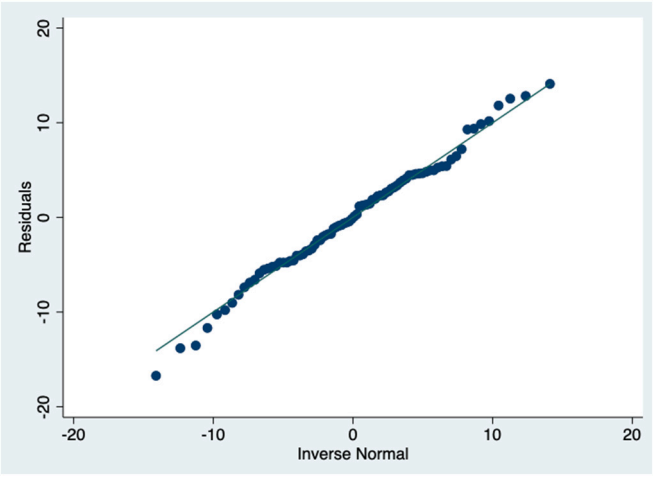


Fig. 1. Q-Q Plot.

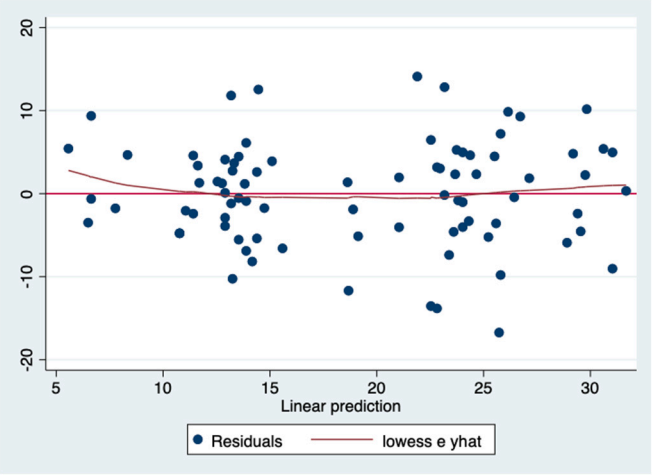


Fig. 2. Linearity test.

were also performed by a series of pairwise comparisons to compare the differences by the discipline variables. Cohen's *d* was used for reporting the effect size. For research question three, the research hypothesis was that there are statistically significant differences in participants' RDM skills by their number of years studying in the graduate program, after controlling for the impact of their pre-scores and discipline.

4. Results

4.1. Participants' initial research data management skills

The present study aimed to examine whether online RDM instruction has significant effects on graduate students' RDM skills and whether this impact varies by their discipline and years of research experience at Texas A&M University, a large R1 public university in the southern US. The participants were assessed on their knowledge of basic RDM skills. A total of 84 graduate students were included in the sample. Descriptive statistics of both pre/post-test scores as a function of group (control/intervention) can be viewed in Table 3. The prior ANOVA test on participants' pre-test scores demonstrated no significant difference between the intervention and control group ($F = 0.20$, $p = .66$).

4.2. The overall effect of the online instruction on graduate students' research data management skills

To answer the first research question, a one-way ANCOVA was performed with participants' post-score as the dependent variable and the group assignment as the predictor, along with their department, number of years studying in the current program, and pre-score as covariates. Overall, the results of one-way ANCOVA (See Table 4) revealed that there were statistically significant differences in participants' RDM skills between experimental and control groups ($F = 59.07$, $p < .0001$) with a large effect size ($\eta^2 = 0.43$; Pierce, Block, & Aguinis, 2004) holding the covariates constant, and the entire model accounted for about 56.75% of variance explained by the group variables ($R^2 = 0.5675$). The descriptive statistics showed that after receiving the online RDM instruction, participants' adjusted mean in the intervention group reported 11.12 points higher than the participants in the control group on their RDM skill scores ($p < .001$). Fig. 3 also describes the posttest comparison between the experimental and control groups.

Table 3
Descriptive statistics of the participants' pre/post scores.

Groups	Pre-Score			Post-Score		
	N	M_{adj}	SD	N	M_{adj}	SD
Control Group	44	12.25	6.42	44	13.10	5.81
Intervention Group	40	12.83	5.40	40	25.53	8.40
Total	84	12.54	5.93	84	19.01	9.47

Table 4
The results of ANCOVA.

Sources	SS	df	MS	F	η^2
Group	2468.47	1	2468.47	59.07***	0.43
Discipline	640.82	3	213.61	5.11**	0.16
Year of study	1.44	1	1.44	0.03	
Pre-score	113.49	1	113.49	2.72	
Residual	3217.58	77	41.79		

Note: *$p < .05$, **p < .01, ***$p < .001$.

4.3. The effect of the online instruction on graduate students' research data management skills varies by their disciplinary focus

To answer the second research question and examine the differences in graduate students' RDM skills by their disciplinary focus, the same one-way ANCOVA was employed with participants' post-score as the dependent variable and their disciplinary focus as the predictor, along with group assignment, number of years studying in the current program, and pre-score as covariates. Descriptive statistics of participants' post-test scores by the disciplinary focus can be viewed in Table 5.

Overall, the results of one-way ANCOVA (See Table 4) revealed that there were statistically significant differences in participants' RDM skills by disciplinary focus ($F = 5.11$, $p < .01$) with a large effect size ($\eta^2 = 0.16$). Fig. 4 described a posttest comparison of the marginal means between the four disciplinary focuses. The plot demonstrated that students from TTED had the highest post-scores compared with students from other disciplinary focuses, and students from EAHR scored lowest on average in the post-assessment.

A series of follow-up tests were performed by pairwise comparisons to examine the difference between the disciplinary focus (See Table 6). Since there were multiple statistical tests, Bonferroni adjustment was used to prevent data from incorrectly appearing to be statistically significant. When Bonferroni correction was applied, the alpha value needed to be lowered to account for the number of comparisons being performed. The follow-up tests revealed that, by holding other factors constant, participants in TTED reported significantly higher scores than the participants in EPSY on their posttest of RDM skills ($t = 3.53$, $p < .008$) with a large effect size (Cohen'$d = 0.89$). The mean differences between the two groups were 8.89 points. Additionally, the follow-up tests showed that participants in TTED reported significantly higher

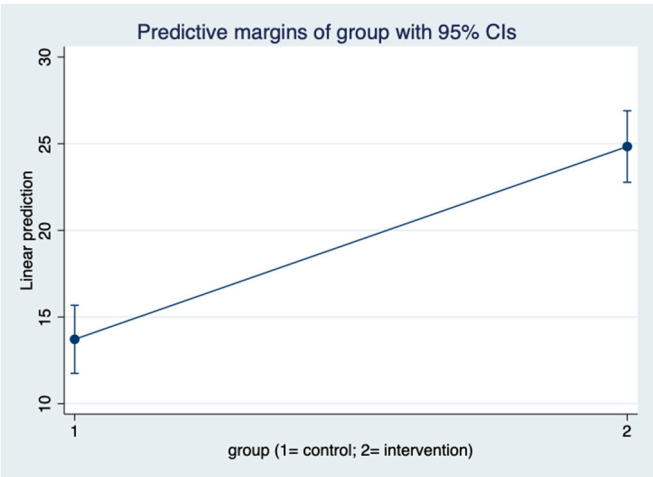


Fig. 3. Posttest comparison between the intervention group and control group.

Table 5
Descriptive statistics of participants' posttest scores by department.

Disciplinary focus	N	M_{adj}	SD
EPSY	47	18.20	8.26
TTED	15	23.93	7.68
EAHR	8	12.78	5.14
HLKN	14	20.04	11.20
Total	84	18.69	9.47

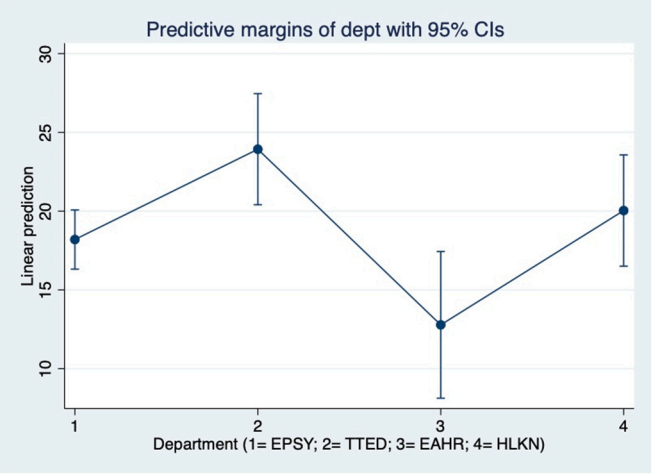


Fig. 4. Posttest comparison of the marginal means between the four disciplines.

Table 6
Follow-up pairwise comparison by department.

Contrasts	Differences	t	p	Cohen's d
TTED vs EPSY	8.89	3.53	0.004*	0.89
TTED vs EAHR	17.06	4.58	0.000*	1.72
EAHR vs EPSY	−8.17	−2.51	0.084	
EAHR vs HLKN	9.13	2.42	0.106	
HLKN vs EPSY	0.96	0.37	1.000	
HLKN vs TTED	−7.93	−2.51	0.084	

Note: Bonferroni correction was used for the department comparisons.
* $p < .008$ (The alpha value was lowered to account for the number of comparisons being performed, in our case it was reduced from 0.05 to 0.008.)

scores than the participants in EAHR on their posttest of RDM skills ($t = 4.58$, $p < .008$) with a large effect size (Cohen'$d = 1.72$). The students from TTED scored 17.06 points on average higher on the marginal mean than those from EAHR. However, no statistically significant differences were found between the comparisons between the other disciplinary focuses (See Fig. 4).

4.4. The effect of the online instruction on graduate students' research data management skills varies by years of study in their program

The same one-way ANCOVA was used with participants' post-score as the dependent variable and the number of years studying in the current program as the predictor, along with group assignment, disciplinary focus, and pre-score as covariates in order to answer the third research question and examine the differences in graduate students' RDM skills by their years studying in the graduate program. Overall, the results of one-way ANCOVA revealed that there were no statistically significant differences in participants' RDM skills by years of studying in the current program ($F = 0.03$, $p = .85$). Descriptive statistics of participants' post-test scores by years of study can be viewed in Table 7. Even though participants' RDM skills showed no significant difference between years of study in the graduate program, students in the program

Table 7
Descriptive statistics of participants' posttest scores by year of studying.

Year of studying	N	M_{ajd}	SD
2 years or less	54	18.83	9.52
More than 2 years	30	19.33	9.53
Total	84	19.01	9.47

for more than two years scored 0.50 higher than the students in the program less than two years.

5. Discussion

In order to satisfy the needs at various steps of the research data life cycle, in the past decade academic libraries have put more emphasis on new initiatives and resources for RDM services, such as data management plan consultations, help with data sharing, and facilitation for open access data (Chabot et al., 2016). Among the RDM services provided by academic libraries, RDM training is a core service (Si et al., 2015). For instance, the University of Edinburgh, the University of Minnesota, and the California Digital Library provide workshops on RDM (Conrad, Shorish, Whitmire, & Hswe, 2017). Previous literature (Xu, 2022; Xu et al., 2022) indicated that most training and instruction in the RDM field targeted faculty, researchers, and librarians. RDM training for graduate students is in great need since it is vital to inculcate good data management skills in them before they become the next generation of faculty and industry researchers (Thielen et al., 2017).

The current study is an empirical study that designed and implemented an intervention for RDM training for graduate students in social science. Empirical studies are important to the development of librarianship practice and research (Crawford & Feldt, 2007). Previous research (Xu, 2022; Xu et al., 2022) demonstrated that needs assessment studies comprised a large portion of RDM training research and showed a great need for RDM training. The RDM field lacks an evidence base of empirical studies on the effectiveness of RDM training. According to Connaway, Radford, and Powell (2016) and VanScoy and Fontana (2016), empirical studies can inform the development of RDM services and provide evidence to support academic libraries' value. Unlike the existing studies that were conducted in non-social science disciplines which often use student feedback as an evaluation of the training effectiveness (Johnston & Jeffryes, 2014; Fong & Wang, 2015), and lack effect size reporting for statistical significance tests (Eaker, 2014), the current study filled this research gap and provided statistical evidence for intervention in RDM training. This present study revealed that the RDM training offered had a great impact on graduate students' RDM skills, with a large effect size ($\eta^2 = 0.43$; Pierce et al., 2004). For most of the participants, this might have been the first time that they received formal RDM training. The study's assessment focused on the knowledge test and the research team conducted the assessment immediately after the participants received the instruction. Future studies could be conducted to assess the study effect after several months or even years and to explore if the participants applied what they learned about RDM to their research.

Furthermore, the current study found that participants' RDM skills varied by disciplinary focus, with a large effect size ($\eta^2 = 0.16$). Specifically, graduate students from TTED had higher post-scores on RDM skills than graduate students from EPSY and EAHR. Previous literature noted that disciplinary differences existed in data management issues (Connaway & Dickey, 2009; Digital Curation Center, 2010) but seldom provided statistical evidence that researchers' RDM skills varied by discipline. This leads to future research investigating the disciplinary differences between graduate students' RDM skills and examining the underlying reasons. However, this present study contained 84 participants, whose sample size might not be big enough to detect differences between four different disciplines, and researchers might need to be cautious when interpreting the findings. Further research can also

attempt to recruit a larger sample size to investigate the differences between disciplines.

The authors of the present study hypothesized that graduate students' understanding of RDM might vary according to their years in the graduate program, which was not supported by this intervention study. However, even though a statistical significant difference in participants' RDM skills between years of study in the graduate program was not detected, students in the program for more than two years scored 0.50 higher than those in the program for less than two years. Students' understanding of RDM might primarily depend on their research experience (Doucette & Fyfe, 2013; Gurat, 2018), which might indicate that years of study in the graduate program might not be positively correlated with their research experience. Future studies could explore more potential reasons for this.

6. Conclusions and future study

While the role of RDM training for graduate students becomes increasingly central to their research and study, training and instruction related to RDM for social science graduate students is still limited. In a recently conducted scoping review, Xu et al. (2022) reported that only 7.26% of studies in the literature focused their RDM training on graduate students, and most of the subjects are centered in STEM subject areas. That is, RDM training is still ignored in the social science discipline. Therefore, it is important to provide RDM training to graduate students in social science disciplines and to understand how the instruction about RDM affects these students' skills in RDM. By testing the effect of a four-hour RDM instruction on graduate students in the social science field with a one-way ANCOVA, the present study showed that online RDM instruction indeed has a significant effect on the improvement of students' knowledge and skills in RDM. Additionally, the positive effect of online instruction for RDM found in this study provides evidence to support RDM instruction in multiple contexts (e.g., online, face-to-face, hybrid), as a prior review (Oo et al., 2021) identified face-to-face instruction as the typical context of RDM instruction.

However, as a pilot study, this study used a self-developed instrument which was designed from existing studies to measure students' RDM skills. The RDM field lacks standardized instruments with sophisticated psychometric evaluation to assess students' RDM knowledge and skills. This should prompt researchers to consider developing a standardized instrument for assessing students' RDM skills and the effect of RDM training for future use. Additionally, this present study used the same assessment as the pre- and post-test. The same assessment before and after the intervention can control participants' prior knowledge of RDM and provide better evidence of the effectiveness of the intervention. However, it may not account for non-intervention influences on outcomes and has its own disadvantages (e.g., some students could have remembered or looked up the answers). Future research can attempt to develop the same level assessment with different formats to be used at pre- and post- test. Furthermore, RDM skills require long-term practice, and the present study measured only the knowledge level after a one-shot instruction. Future research needs to focus more on the application of RDM knowledge in real research practice and measure the development of skills longitudinally. Additionally, the results of this study demonstrated that the impact of the RDM instruction varied across the four domains among social science graduate students. This suggests that while RDM training for the broad social science field does impact students' RDM skills, specific program-tailored RDM training within social science disciplines is also important. Future studies should be designed as discipline-based RDM training to maximize students' skills.

Data availability statement

The dataset for this study has been published in Texas Data repository at https://dataverse.tdl.org/dataset.xhtml?persistentId=doi:10.18738/T8/NQLIJE.

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Declaration of Competing Interest

We declare that we have no conflicts of interest.

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